

Fused Routed Decode Attention (compute_decode_attention_static, mx.compile)

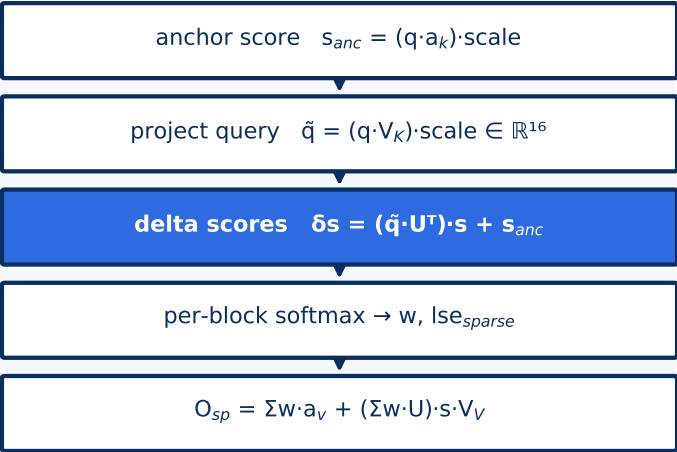
Route top-K by EXACT residual-key score $\rightarrow$ low-rank (approximation) + exact residuals + recency $\rightarrow$ LSE merge

query q [H, d]

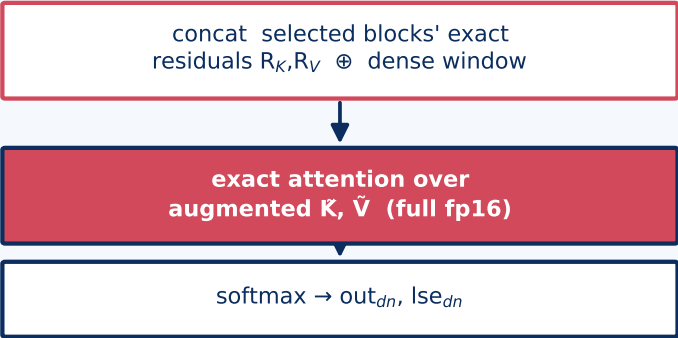
outer: $\rho_b = \max(q \cdot a_k, \max_j q \cdot R_{K,j}) \cdot \text{scale} \rightarrow \text{keep top-K} = 16 \text{ blocks}$

exact $q \cdot k$ over
anchor + residual keys

Low-rank branch (selected blocks)



Exact branch (residuals + recency)



verbatim recall of needle tokens

$O(K \cdot r \cdot B)$ — no $O(B \cdot d)$ decomposition

flash-style LSE merge
 $m = \max(lse_{sp}, lse_{dn})$
 $out = (e^{-m} O_{sp} + e^{-m} O_{dn}) / \Sigma$

NaN/inf-guarded so an empty compressed set contributes exactly zero weight